



Lab Report: Thin Lenses and Image Formation

Student Name: _____

Date: _____

Grade/Section: _____

1. Objective

To investigate the relationship between object distance (d_o), image distance (d_i), and focal length (f) for convex and concave lenses and to determine magnification.

2. Materials

- DARAWA Lenses Simulation (Desktop/AR Mode)
- Virtual Optical Bench, Light Source, and Screen

3. Theoretical Background

The behavior of thin lenses is governed by the Thin Lens Equation:

$$\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i}$$

Magnification (M) is given by:

$$M = \frac{h_i}{h_o} = -\frac{d_i}{d_o}$$

4. Data Collection

Record your measurements from the simulation in the table below.

Trial	Lens Type	Obj. Dist (d_o)	Img. Dist (d_i)	Focal Length (f)	Calc. $1/f$
1	Convex				
2	Convex				
3	Convex				
4	Concave				

5. Analysis Questions

1. Using Trial 1, calculate the focal length using the lens equation. Show your work.
2. Describe the image characteristics (Real/Virtual, Inverted/Upright) when the object is placed at $2f$.



3. For the concave lens (Trial 4), is the image distance positive or negative? Explain what this implies about the image.